

Lecture Note in Calculus B - Final

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6.2-Volumes

Def. [volume]:

Let “S” be solid that lies between $x = a$ and $x = b$. If the cross-sectional area of S in the plane P_x , through x perpendicular to x – axis, is $A(x)$, where A is a continuous function, then the volume of S is:

$$V = \int_a^b A(x) dx \text{ (or } dy) = \int_a^b \pi r^2 dx \text{ (or } dy)$$

Note:

- If the rotation axis is the ***x-axis*** then we will integrate with respect to x

$$V = \int_a^b \pi r^2 dx$$

- If the rotation axis is the ***y-axis*** then we will integrate with respect to y

$$V = \int_a^b \pi r^2 dy$$

Examples

1. Find the volume of the solid obtained by rotating about x -axis the region under the curve $y = \sqrt{x}$ from 0 to 1 .

sol:

$$\begin{aligned} V &= \int_a^b \pi r^2 dx = \int_0^1 \pi r^2 dx \\ &= \int_0^1 \pi (\sqrt{x})^2 dx = \int_0^1 \pi x dx \\ &= \left[\pi \frac{x^2}{2} \right]_0^1 = \frac{\pi}{2} \end{aligned}$$

2. The region R enclosed by curves $y = x$ and $y = x^2$ is rotated about:

a- x - axis

Sol:

To get the intersection points: $y = y$

$$\rightarrow x = x^2 \rightarrow x^2 - x = 0 \rightarrow x(x - 1) = 0$$

$$\therefore x = 0, x = 1$$

$$x > x^2 > 0 \quad \forall x \in [0, 1]$$

$$V = \int_a^b \pi r^2 dx = \pi \int_0^1 [(x - 0)^2 - (x^2 - 0)^2] dx$$

$$= \pi \int_0^1 (x^2 - x^4) dx = \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \frac{2\pi}{15}$$

b- $x = -1$

Sol:

$$x = y, x = \sqrt{y}$$

$$\sqrt{y} > y > -1 \quad \forall y \in [0,1]$$

$$V = \int_a^b \pi r^2 dy = \pi \int_0^1 \left[\left(\sqrt{y} - (-1) \right)^2 - \left(y - (-1) \right)^2 \right] dy$$

$$V = \frac{\pi}{2}$$

3. **Set up an integral** to find the volume of the solid generated by rotating the region bounded by the graphs $y = -x^2 + 4$ and $x = y - 2$ about the line $y = -1$

Sol:

$$\begin{aligned}y &= y \\-x^2 + 4 &= x + 2 \\x^2 + x - 2 &= 0 \\\therefore x &= -2, x = 1 \\(-x^2 + 4) &> (x + 2) > -1, \forall x \in [-2, 1] \\V &= \int_a^b \pi r^2 dx \\&= \pi \int_{-2}^1 \left[((-x^2 + 4) + 1)^2 - ((x + 2) + 1)^2 \right] dx\end{aligned}$$

4. Let R be the region bounded by graphs of $y + x^2 - 2x = 0$ and $y + 2x = 0$. **set up an integral** to find the volume of the solid generated by rotating R about $y = 1$

Sol:

$$\begin{aligned}y &= y \\2x - x^2 &= -2x \\x^2 - 4x &= 0 \\\therefore x &= 0, x = 4 \\1 &> 2x - x^2 > -2x, \forall x \in [0, 4] \\V &= \int_a^b \pi r^2 dx \\&= \pi \int_0^4 \left[(1 - (-2x))^2 - (1 - (2x - x^2))^2 \right] dx\end{aligned}$$

8.1 + 8.2-Arc length + area of surface

Def.:

- If f' cont. on $[a, b]$, then the length of the curve $y = f(x)$,

$$a \leq x \leq b \text{ is: } L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

- If $x = g(y)$, $c \leq y \leq d$ such that $g'(y)$ is cont. on $[c, d]$,

$$\text{then: } L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Summery [8.1 + 8.2]:

• If $y = f(x)$, $a \leq x \leq b$:

- The length of the curve: $L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

- Area of the curve if the curve is rotated:

$$\text{About } x - \text{axis: } S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

$$\text{About } y - \text{axis: } S = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx..$$

• If $x = g(y)$, $c \leq y \leq d$:

- The length of the curve: $L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

- Area of the curve if the curve is rotated:

$$\text{About } x - \text{axis: } S = \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy.$$

$$\text{About } y - \text{axis: } S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy..$$

Examples

1. Find the arc length of $f(x) = \ln(\sec x)$, $0 \leq x \leq \frac{\pi}{4}$.

Sol:

$$\begin{aligned} L &= \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_0^{\frac{\pi}{4}} \sqrt{1 + \left(\frac{\sec x \tan x}{\sec x}\right)^2} dx \\ &= \int_0^{\frac{\pi}{4}} \sqrt{1 + (\tan x)^2} dx = \int_0^{\frac{\pi}{4}} \sec x dx \\ &= [\ln|\sec x + \tan x|]_0^{\frac{\pi}{4}} = \ln(\sqrt{2} + 1) \end{aligned}$$

2. Find the arc length of $y = \int_1^x \sqrt{t^3 - 1} dt$, $1 \leq x \leq 4$.

sol:

$$\begin{aligned} \frac{dy}{dx} &= \left(\sqrt{x^3 - 1} \cdot (1) \right) = \sqrt{x^3 - 1} , [\text{by F.T.C}] \\ L &= \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_1^4 \sqrt{1 + (\sqrt{x^3 - 1})^2} dx \\ &= \int_1^4 x^{3/2} dx = \left[\frac{2}{5} x^{5/2} \right]_1^4 = \frac{62}{5} \end{aligned}$$

3. Find the arc length of $f(x) = x^2 - \frac{1}{8} \ln(x)$, $1 \leq x \leq 3$.

$$\begin{aligned} L &= \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_1^3 \sqrt{1 + \left(2x - \frac{1}{8x}\right)^2} dx \\ &= \int_1^3 \sqrt{1 + 4x^2 - \frac{1}{2} + \frac{1}{64x^2}} dx = \int_1^3 \sqrt{\left(2x + \frac{1}{8x}\right)^2} dx \\ &= \int_1^3 \left(2x + \frac{1}{8x}\right) dx = \left[x^2 + \frac{\ln(x)}{8} \right]_1^3 = \frac{\ln(3)}{8} + 8 \end{aligned}$$

4. Find the area of the surface obtained by rotating the curve $y = \sqrt{4 - x^2}$, $-1 \leq x \leq 1$ about $x - axis$.

Sol:

$$\begin{aligned} S &= \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_{-1}^1 2\pi \sqrt{4 - x^2} \sqrt{1 + \left(\frac{-x}{\sqrt{4-x^2}}\right)^2} dx \\ &= \int_{-1}^1 2\pi \sqrt{4 - x^2} \sqrt{\frac{4}{4-x^2}} dx = 4\pi \int_{-1}^1 dx = 4\pi [x]_{-1}^1 = 8\pi \end{aligned}$$

5. Set up but do not evaluate an integral for the area of the surface obtained by rotating the curve $y = \sec x$, $0 \leq x \leq \frac{\pi}{4}$ about $y - axis$.

sol:

Type equation here.

$$S = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_0^{\frac{\pi}{4}} 2\pi x \sqrt{1 + (\sec x \tan x)^2} dx$$

6. The curve $x = \sqrt{y}$, $1 \leq y \leq 4$ is rotated about $y - axis$. Find the area of the resulting surface.

Sol:

$$\begin{aligned} S &= \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_1^4 2\pi \sqrt{y} \sqrt{1 + \left(\frac{1}{2\sqrt{y}}\right)^2} dy \\ &= \int_1^4 2\pi \sqrt{y} \sqrt{\frac{4y+1}{4y}} dy = \pi \int_1^4 \sqrt{4y+1} dy \\ &= \frac{\pi}{6} \left[(17)^{3/2} - (5)^{3/2} \right] \end{aligned}$$

Exercises-8.1 + 8.2

1. Let $C : y = \frac{x^3}{3} + \frac{1}{4x}$, $x \in [1,2]$

- Find the length of the curve $\rightarrow \left(ans = \frac{59}{24} \right)$

- Find the surface area obtained by rotating C about
 $y - axis \rightarrow \left(ans = \frac{\pi}{2} [15 + \ln(2)] \right)$

2. Let $C : x = \frac{1}{3}(y^2 + 2)^{3/2}$, $y \in [1,2]$

- Find the length of the curve $\rightarrow \left(ans = \frac{10}{3} \right)$

- Find the surface area obtained by rotating C about
 $x - axis \rightarrow \left(ans = \frac{21\pi}{2} \right)$

3. Let $C : y = \cosh x$, $x \in [0,1]$

- Find the length of the curve $\rightarrow \left(ans = \frac{(e - e^{-1})}{2} \right)$

- Find the surface area obtained by rotating C about:

a. $x - axis \rightarrow \frac{\pi}{2} \left[e^2 + 2 - \frac{1}{2e^2} \right]$

b. $y - axis \rightarrow \left[2\pi \left(1 - \frac{1}{e} \right) \right]$

4. Find the length of the curve $y = \ln\left(\frac{e^x+1}{e^x-1}\right)$, $a \leq x \leq b$, $a > 0$
 $\rightarrow (ans = \ln|\sinh(b)| - \ln|\sinh(a)|)$

5. Let $y = \sqrt{x+1}$, $1 \leq x \leq 5$ find area about x - axis
 $\rightarrow (ans = \frac{\pi}{6}(125 - 27))$

6. Find area about y - axis : $y = \frac{x^2}{8} - \ln(x)$, $x \in [1,2]$
 $\rightarrow (ans = \frac{19\pi}{6})$

7. Let $C : y = \ln(\sec x)$, $x \in [0, \frac{\pi}{4}]$ find the length of the
curve
 $\rightarrow (ans = \ln(\sqrt{2} + 1))$

8. Find the arc length of the graph of $f(x) = 2 \int_1^{\sqrt{x}} t\sqrt{t^4 - 1} dt$
from $x = 1$ to $x = 4$
 $\rightarrow (ans = \frac{15}{2})$

8.3-Moments and centers of mass

Def. :

If a region R lies between two curves $y = f(x)$, $y = g(x)$ where $f(x) \geq g(x)$. The **centroid** $(\bar{x}, \bar{y})$ is given by :

$$\bar{x} = \frac{1}{A} \int_a^b x(f(x) - g(x))dx$$

$$\bar{y} = \frac{1}{A} \int_a^b \frac{1}{2} \left[(f(x))^2 - (g(x))^2 \right] dx$$

$$\text{Where } A = \int_a^b [f(x) - g(x)]dx$$

Examples

1. Find the centroid of the region bounded by $y = x$, $y = x^2$.
Sol:

$$y = y$$

$$x = x^2$$

$$x - x^2 = 0$$

$$\rightarrow x = 0, x = 1$$

$$A = \int_a^b [f(x) - g(x)]dx = \int_0^1 [x - x^2]dx = \frac{1}{6}$$

$$\bar{x} = \frac{1}{A} \int_a^b x(f(x) - g(x))dx = 6 \int_0^1 x[x - x^2]dx = \frac{1}{2}$$

$$\bar{y} = \frac{1}{2A} \int_a^b \left[(f(x))^2 - (g(x))^2 \right] dx = 3 \int_0^1 [x^2 - x^4]dx = \frac{2}{5}$$

$$\therefore (\bar{x}, \bar{y}) = \left(\frac{1}{2}, \frac{2}{5} \right)$$

Def. :

Consider a system of n particles with masses $m_1, m_2, \dots, m_n$ located at the points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ in xy -plane. Then we define the moments of the system about:

- The origin : $M = \sum_{i=1}^n m_i x_i$
- The y - axis : $M_y = \sum_{i=1}^n m_i x_i$
- The x - axis : $M_x = \sum_{i=1}^n m_i y_i$

And: $\left(\bar{x} = \frac{M_y}{m}, \bar{y} = \frac{M_x}{m} \right)$ where $m = \sum_{i=1}^n m_i$

2. Find the moments M_x and M_y and the center of mass of the system with: $[m_1 = 6, m_2 = 5, m_3 = 1, m_4 = 4]$ and $[p_1 = (1, -2), p_2 = (3, 4), p_3 = (-3, -7), p_4 = (6, -1)]$.

$$m = \sum_{i=1}^4 m_i = 16$$

$$M_x = \sum_{i=1}^4 m_i y_i = 6(-2) + 5(4) + (-7) + 4(-1) = -3$$

$$M_y = \sum_{i=1}^4 m_i x_i = 6(1) + 5(3) + (-3) + 4(6) = 42$$

$$\left(\bar{x} = \frac{M_y}{m} = \frac{42}{16}, \bar{y} = \frac{M_x}{m} = -\frac{3}{16} \right)$$

$$\therefore \left(\bar{x} = \frac{21}{8}, \bar{y} = \frac{-3}{16} \right)$$

Exercises-8.3

1. Find centroid of region bounded by $y = \sqrt{x-1}$, x -axis, $x = 5 \rightarrow \left(ans = \left(\frac{17}{5}, \frac{3}{4} \right) \right)$

2. Find centroid of $y = \sqrt{1-x^2}$, $y = 0 \rightarrow \left(ans = \left(0, \frac{4}{3\pi} \right) \right)$

3. Find centroid of $y = \sin x$, $y = \cos x$, $x = 0$, $x = \frac{\pi}{4} \rightarrow \left(ans = \left(\frac{1}{\sqrt{2}-1} \left(\frac{\pi}{4} (\sqrt{2}) - 1 \right), \frac{1}{4(\sqrt{2}-1)} \right) \right)$

4. Find centroid $y = 2x^2$, $y = x^3 \rightarrow \left(ans = \left(\frac{6}{5}, \frac{(47)8}{35} \right) \right)$

5. Find centroid $x = y^2$, $x = 4 \rightarrow \left(ans = \left(\frac{12}{5}, 0 \right) \right)$

6. Find centroid $x = y^2$, $y = x^2 \rightarrow \left(ans = \left(\frac{9}{20}, \frac{1}{4} \right) \right)$

7. Find center of mass $y = \sec^2 x$, x -axis, y -axis, $x = \frac{\pi}{4} \rightarrow \left(ans = \left(\frac{\pi - 2 \ln(2)}{4}, \frac{2}{3} \right) \right)$

8. $x = 5 - y^2$ and $x = 0 \rightarrow (ans = (2, 0))$

9. $y = \frac{1}{x}$, $y = 0$, $x = 1$, $x = 2 \rightarrow \left(ans = \left(\frac{1}{\ln(2)}, \frac{1}{4 \ln(2)} \right) \right)$

10.1 + 10.2-Parametric equations

Def.:

Let f and g be cont. functions on $[a, b]$. The equations $x=f(t)$, $y=g(t)$, $t \in [a, b]$ are called **parametric equations** of the curve C . C consist of all points $P(t) = (f(t), g(t))$.

C has **initial point** $P(a) = (f(a), g(a))$ and

terminal point $P(b) = (f(b), g(b))$.

- If $P(a) = P(b)$ we call the curve **C closed curve**.
- C has a direction pointing from $P(t_1)$ to $P(t_2)$, $t_1 < t_2$.

Examples

1. Sketch and identify the curve of parametric eq.

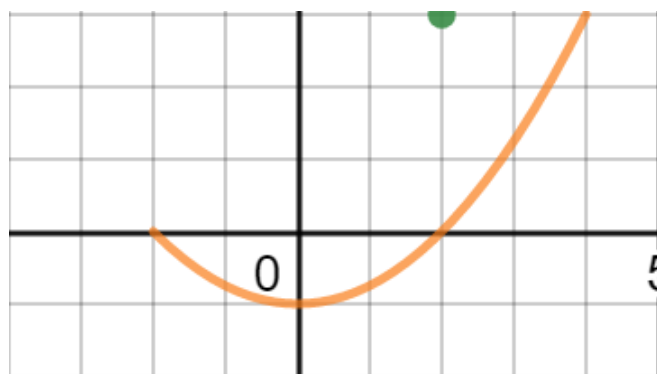
$$x = 2t, y = t^2 - 1, -1 \leq t \leq 2$$

t	-1	0	1	2
x	-2	0	2	4
y	0	-1	0	3

The curve is **not closed** since $I.P \neq T.P$

$$\because x = 2t \rightarrow t = \frac{x}{2}$$

$$\therefore y = \frac{x^2}{4} - 1, \text{ which is parabola}$$



2. Describe the motion of a particle with position (x, y) as t varies in the given interval :

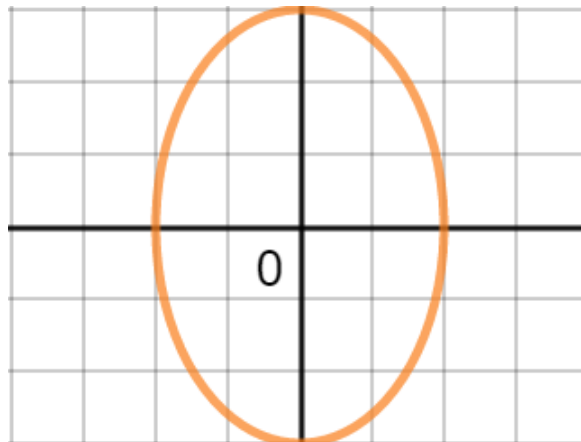
$$x = 2 \sin t, \quad y = 3 \cos t, \quad 0 \leq t \leq 2\pi$$

t	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
x	0	2	0	-2	0
y	3	0	-3	0	3

The curve is **closed I.P = T.P**

We know $\sin^2 t + \cos^2 t = 1$

$\therefore \frac{x^2}{4} + \frac{y^2}{9} = 1$, which is ellipse



3. $(x^2 - 2x) + (y^2 - 4y) = 0$

$$x^2 - 2x + 1 - 1 + y^2 - 4y + 4 - 4 = 0$$

$$(x - 1)^2 - 1 + (y - 2)^2 - 4 = 0$$

$$(x - 1)^2 + (y - 2)^2 = 5$$

Which is **circle** with $r = \sqrt{5}$

$$x = \sqrt{5} \cos t + 1$$

$$y = \sqrt{5} \sin t + 2$$

SUMMARY [10.1 and 10.2]

Let $C : x = g(t)$, $y = h(t)$, $a \leq t \leq b$

- $y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

- *Horizontal tangent line :*

$$\frac{dy}{dt} = 0 \quad , \quad \frac{dx}{dt} \neq 0$$

- *Vertical tangent line :*

$$\frac{dx}{dt} = 0 \quad , \quad \frac{dy}{dt} \neq 0$$

- *The length of curve :*

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

- *Area about :*

$$\text{- } x\text{-axis : } S = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\text{- } y\text{-axis : } S = \int_a^b 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

- $y'' = \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt}y'}{\frac{dx}{dt}}$

4. Let $C : x = \ln(\sqrt{1+t^2})$, $y = \tan^{-1}(t)$, $t \in [0,1]$
- Find eq. of tangent line at ($t = 1$).
 - Find the length of the curve.

Sol:

$$y - y_0 = m(x - x_0)$$

$$x_0 = x(t = 1) = \ln(\sqrt{2})$$

$$y_0 = y(t = 1) = \frac{\pi}{4}$$

$$m = y'|_{t=1} = 1$$

$$\therefore \text{eq. of T.L : } y - \frac{\pi}{4} = (x - \ln(\sqrt{2}))$$

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^1 \sqrt{\left(\frac{t}{1+t^2}\right)^2 + \left(\frac{1}{1+t^2}\right)^2} dt$$

$$= \int_0^1 \sqrt{\frac{t^2+1}{(1+t^2)^2}} dt = \int_0^1 \frac{1}{\sqrt{1+t^2}} dt$$

$$\text{let } t = \tan \theta \quad , \quad dt = \sec^2 \theta$$

$$L = \int_0^{\pi/4} \frac{\sec^2 \theta}{\sec \theta} d\theta = \int_0^{\pi/4} \sec \theta d\theta$$

$$= [\ln|\sec \theta + \tan \theta|]_0^{\frac{\pi}{4}}$$

$$= \ln|1 + \sqrt{2}|$$

5. Let C given by parametric eq.'s : $x = t^2$, $y = t^3 - 3t$
 - Show that C has two tangent at $(3,0)$ and find the eq. of T.L .
 Sol:

$$\text{at } (3,0) \rightarrow y = 0 \rightarrow t^3 - 3t = 0 \rightarrow t = \pm\sqrt{3}$$

$$m_1 = \left(\frac{3t^2 - 3}{2t} \right)_{t=\sqrt{3}} = \sqrt{3}$$

$$\therefore y - 0 = \sqrt{3} (x - 3)$$

$$m_2 = \left(\frac{3t^2 - 3}{2t} \right)_{t=-\sqrt{3}} = -\sqrt{3}$$

$$\therefore y - 0 = -\sqrt{3} (x - 3)$$

- Find the points on C at which the tangent (horizontal-vertical) .

$$\text{V.T.L : } \frac{dx}{dt} = 0 \rightarrow 2t = 0 \rightarrow t = 0 , \text{ point} = (0,0)$$

$$\text{H.T.L : } \frac{dy}{dt} = 0 \rightarrow 3t^2 - 3 = 0 \rightarrow t = \pm 1$$

$$t = 1 \rightarrow (1, -2)$$

$$t = -1 \rightarrow (1, 2)$$

- Determine whether C is CU or CD .

$$y'' = \frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt} y'}{\frac{dx}{dt}}$$

$$y'' = \frac{\frac{3}{2} \left(1 + \frac{1}{t^2} \right)}{2t}$$

$$\text{if } t > 0 , \text{ CU}$$

$$\text{if } t < 0 , \text{ CD}$$

Let C given by : $x = \cos t + \sin(2t)$

$y = \sin t + \cos(2t)$ Find eq.'s of T.L at $t = \pi$

Sol:

$$y - y_0 = m(x - x_0)$$

$$x_0 = x(t = \pi) = -1$$

$$y_0 = y(t = \pi) = 1$$

$$m = y'|_{t=\pi} = \frac{-1}{2}$$

$$\therefore \text{eq. of T.L : } y - 1 = \frac{-1}{2}(x + 1)$$

6. Find the length of the curve C :

$$x = a \cos t, \quad y = a \sin t, \quad 0 \leq t \leq 2\pi$$

Sol:

$$\begin{aligned} L &= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \\ &= \int_0^{2\pi} \sqrt{a^2 \sin^2 t + a^2 \cos^2 t} dt \\ &= \int_0^{2\pi} \sqrt{a^2(\sin^2 t + \cos^2 t)} dt = a \int_0^{2\pi} dt \\ &= 2\pi a \end{aligned}$$

7. Show that the surface area of sphere of radius r is $4\pi r^2$.

Consider the rotation of the semi-circle about x-axis

$$0 \leq t \leq \pi, \quad x = r \cos t, \quad y = r \sin t$$

$$\begin{aligned} S &= \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \\ &= 2\pi \int_0^\pi r \sin t \sqrt{r^2 \sin^2 t + r^2 \cos^2 t} dt \\ &= 2\pi \int_0^\pi r \sin t \sqrt{r^2(\sin^2 t + \cos^2 t)} dt \\ &= 2\pi \int_0^\pi r^2 \sin t dt \\ &= 2\pi r^2 [-\cos t]_0^\pi = 4\pi r^2 \end{aligned}$$

H.W:

1. Identify the curve by finding cartesian coordinates:

C: $x = 2 + \cos t$, $y = -3 + \sin t$

2. Identify the curve by finding polar coordinates :

a) $x^2 + y^2 = a^2$

b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

c) $x^2 - 2x + y^2 - 4y = 0$

3. Let C: $x = \sin^{-1}(t)$, $y = \sec^{-1}(2t)$

Find the equation of T.L at $t=1$

Areas

We know that the area under a curve $y = F(x)$ from a to b is $A = \int_a^b F(x) dx$, where $F(x) \geq 0$. If the curve is traced out once by the parametric equations $x = f(t)$ and $y = g(t)$, $\alpha \leq t \leq \beta$, then we can calculate an area formula by using the Substitution Rule for Definite Integrals as follows:

The limits of integration for t are found as usual with the Substitution Rule. When $x = a$, t is either α or β . When $x = b$, t is the remaining value.

$$A = \int_a^b y dx = \int_{\alpha}^{\beta} g(t) f'(t) dt \quad \left[\text{or} \quad \int_{\beta}^{\alpha} g(t) f'(t) dt \right]$$

EXAMPLE 3 Find the area under one arch of the cycloid

$$x = r(\theta - \sin \theta) \quad y = r(1 - \cos \theta)$$

(See Figure 3.)

SOLUTION One arch of the cycloid is given by $0 \leq \theta \leq 2\pi$. Using the Substitution Rule with $y = r(1 - \cos \theta)$ and $dx = r(1 - \cos \theta) d\theta$, we have

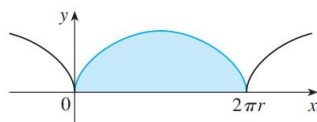


FIGURE 3

The result of Example 3 says that the area under one arch of the cycloid is three times the area of the rolling circle that generates the cycloid (see Example 10.1.7). Galileo guessed this result but it was first proved by the French mathematician Roberval and the Italian mathematician Torricelli.

$$\begin{aligned} A &= \int_0^{2\pi r} y dx = \int_0^{2\pi} r(1 - \cos \theta) r(1 - \cos \theta) d\theta \\ &= r^2 \int_0^{2\pi} (1 - \cos \theta)^2 d\theta = r^2 \int_0^{2\pi} (1 - 2 \cos \theta + \cos^2 \theta) d\theta \\ &= r^2 \int_0^{2\pi} \left[1 - 2 \cos \theta + \frac{1}{2}(1 + \cos 2\theta) \right] d\theta \\ &= r^2 \left[\frac{3}{2}\theta - 2 \sin \theta + \frac{1}{4} \sin 2\theta \right]_0^{2\pi} \\ &= r^2 \left(\frac{3}{2} \cdot 2\pi \right) = 3\pi r^2 \end{aligned}$$

Exercises-10.1 + 10.2

1. Let $C : x = e^t \cos t$, $y = e^t \sin t$, $t \in [0, 2\pi]$

- Find eq. of T.L at $t = \frac{\pi}{2} \rightarrow (ans: y - e^{\pi/2} = -x)$

- Find pts at which T.L is:

i. Vertical $\rightarrow (ans: (\frac{e^{\pi/4}}{\sqrt{2}}, \frac{e^{\pi/4}}{\sqrt{2}}), (\frac{-e^{5\pi/4}}{\sqrt{2}}, \frac{e^{-5\pi/4}}{\sqrt{2}}))$

ii. Horizontal $\rightarrow (ans: (\frac{e^{-3\pi/4}}{\sqrt{2}}, \frac{e^{3\pi/4}}{\sqrt{2}}), (\frac{e^{7\pi/4}}{\sqrt{2}}, \frac{-e^{7\pi/4}}{\sqrt{2}}))$

- Find the length of the curve $\rightarrow (ans: \sqrt{2} (e^{2\pi} - 1))$

- Find the area of surface obtained by rotating C about y-axis
 $\rightarrow (ans: \frac{2\pi\sqrt{2}}{5} (2e^{4\pi} - 2))$

2. Let $C : x = t - \sin t$, $y = 1 - \cos t$, $t \in [0, 2\pi]$

- Determine if C is CU or CD at $t = \pi \rightarrow (ans: CD)$

- Find the area of surface obtained by rotating C about x-axis
 $\rightarrow (ans: \frac{64\pi}{3})$

3. Let $C : x = 2e^t - 3t$, $y = e^t + 5t$, find pts on curve C at which the T.L is parallel to the line $y = 7x + 1 \rightarrow$
(ans: $(4 - \ln 8, 2 + \ln 32)$)

4. Let $C : x = e^t - t$, $y = 2e^t$, $t \in \mathbb{R}$

- Find pts at which T.L is:

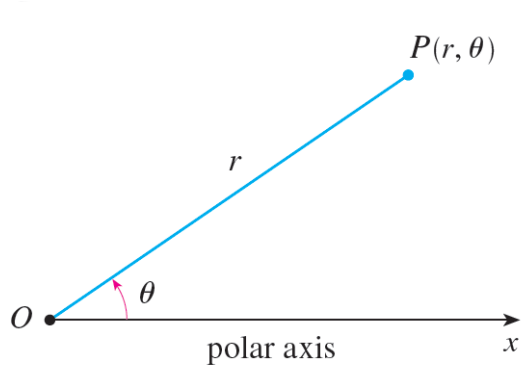
iii. Vertical $\rightarrow (ans: (1,2))$

iv. Horizontal $\rightarrow (ans: no\ solution)$

- Find an interval at which C is CU or CD and find inflection pt if any $\rightarrow (ans: CU\ on\ (-\infty, 0)$, $CD\ on\ (0, \infty)$, $inf.\ pt\ (1,2))$

5. Let $C : x = \cos^3 t$, $y = \sin^3 t$, $t \in \left[0, \frac{\pi}{2}\right]$, Find area of surface generated by rotating C about x-axis $\rightarrow \left(ans: \frac{6\pi}{5}\right)$

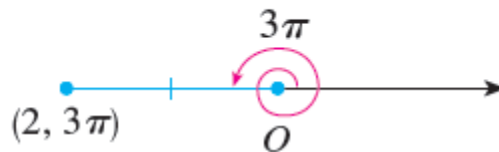
10.3-Polar coordinates



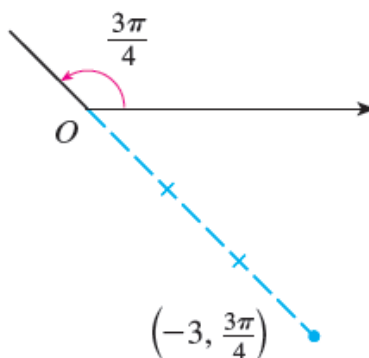
 r and θ are called polar coordinates.

Examples

1. Plot the point whose polar coordinates are $(2, 3\pi)$.



2. $\left(-3, \frac{3\pi}{4}\right)$



Remark:

عندما نعوض وتكون r سالبة يجب أن نجعلها موجبة بنفس قيمتها (أي نغير اشاراتها فقط) وذلك من خلال إضافة π على الزاوية θ

في الرسم بنظام ال polar دائما نتجه باتجاه الزاوية θ بمقدار قيمة r

في الرسم بنظام ال polar عندما ننزل للأسفل ليس لنا علاقة بالسالب

عندما تكون الزاوية θ نتحرك بمقدار قيمة 90°

$$(0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi)$$

عندما تكون الزاوية 2θ نتحرك بمقدار قيمة 45°

$$(0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \dots)$$

$\sin(n\theta) = n$ petals

$\cos(n\theta) = 2n$ petals

■ Symmetry

When we sketch polar curves it is sometimes helpful to take advantage of symmetry. The following three rules are explained by Figure 14.

- (a) If a polar equation is unchanged when θ is replaced by $-\theta$, the curve is symmetric about the polar axis.
- (b) If the equation is unchanged when r is replaced by $-r$, or when θ is replaced by $\theta + \pi$, the curve is symmetric about the pole. (This means that the curve remains unchanged if we rotate it through 180° about the origin.)
- (c) If the equation is unchanged when θ is replaced by $\pi - \theta$, the curve is symmetric about the vertical line $\theta = \pi/2$.

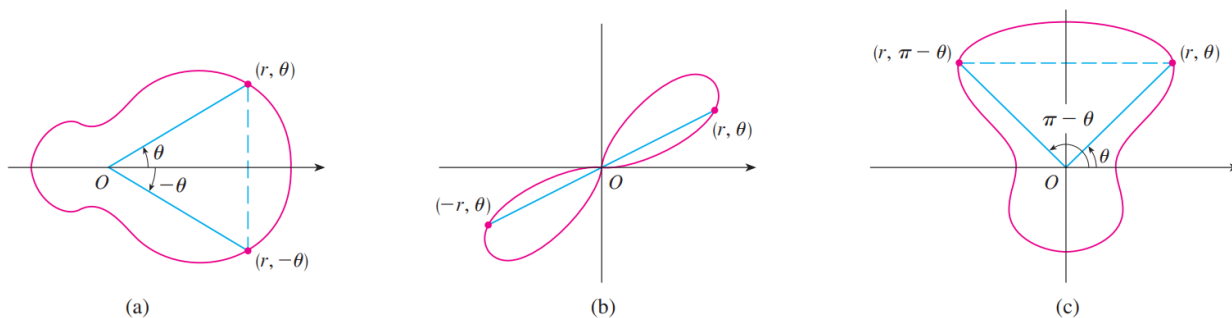
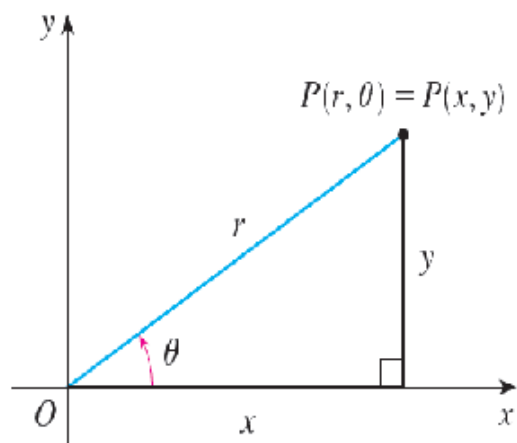


FIGURE 14

The curves sketched in Examples 6 and 8 are symmetric about the polar axis, since $\cos(-\theta) = \cos \theta$. The curves in Examples 7 and 8 are symmetric about $\theta = \pi/2$ because $\sin(\pi - \theta) = \sin \theta$ and $\cos 2(\pi - \theta) = \cos 2\theta$. The four-leaved rose is also symmetric about the pole. These symmetry properties could have been used in sketching the curves. For instance, in Example 6 we need only have plotted points for $0 \leq \theta \leq \pi/2$ and then reflected about the polar axis to obtain the complete circle.

Relation between cartesian coordinates and polar coordinates



$$\cos \theta = \frac{x}{r} \rightarrow x = r \cos \theta$$

$$\sin \theta = \frac{y}{r} \rightarrow y = r \sin \theta$$

$$\tan \theta = \frac{y}{x}$$

$$r^2 = x^2 + y^2$$

3. Let $r = 8 \cos \theta - 6 \sin \theta$, find the cartesian coordinates.

Sol:

Multiply two sides by “r”

$$r^2 = 8 r \cos \theta - 6 r \sin \theta$$

$$x^2 + y^2 = 8x - 6y$$

$$x^2 - 8x + y^2 + 6y = 0$$

$$(x - 4)^2 + (y + 3)^2 = 25$$

Circle with center $(4, -3)$ and $r = 5$

Circles :-

- $r = a$, $a > 0$
- $r = \pm a \cos \theta$
- $r = \pm a \sin \theta$

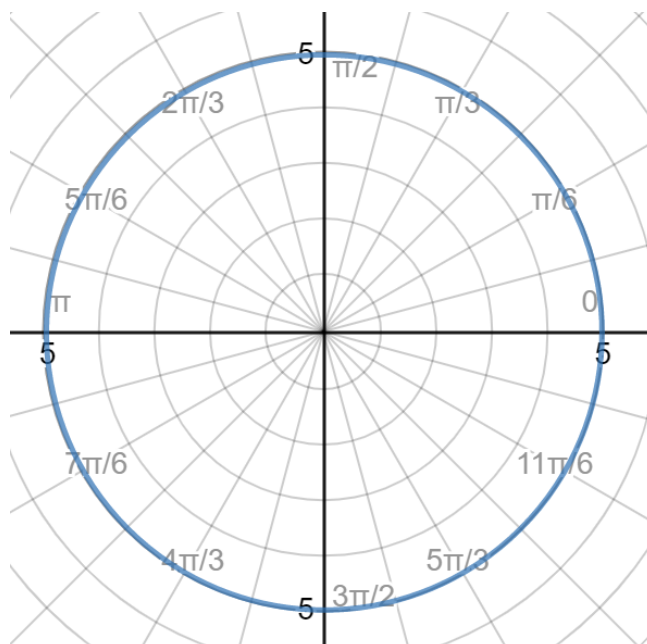
Cardioids :-

- $r = a \pm a \cos \theta$
- $r = a \pm a \sin \theta$

Examples Sketch the following:

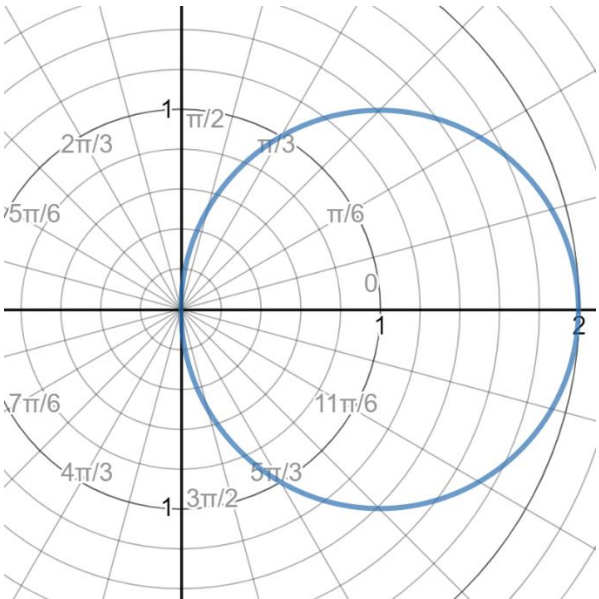
4. $r = 5$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	5	5	5	5	5



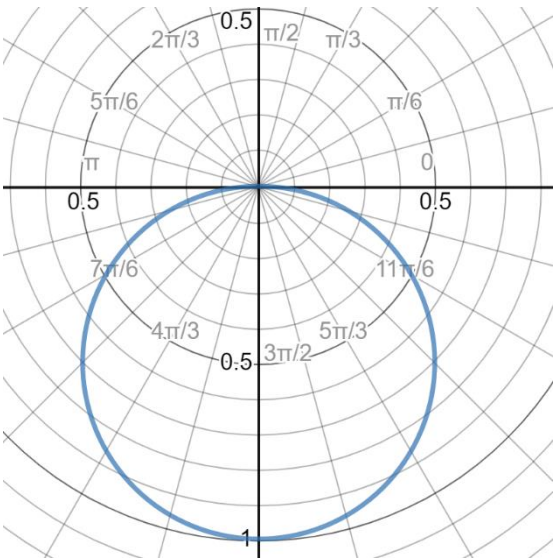
5. $r = 2 \cos \theta$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	2	0	-2	0	2



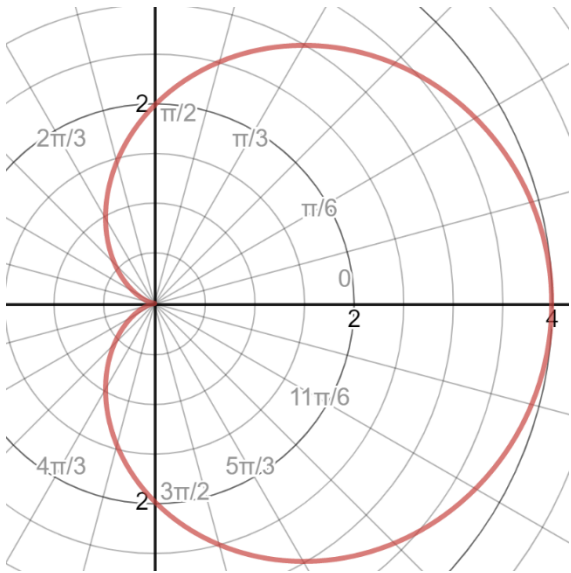
6. $r = - \sin \theta$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	0	-1	0	1	0



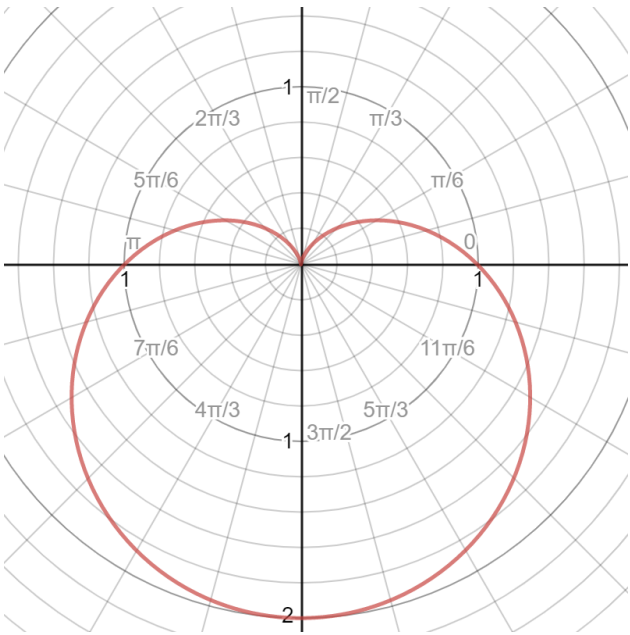
7. $r = 2 + 2 \cos \theta$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	4	2	0	2	4



8. $r = 1 - \sin \theta$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	1	0	1	2	1



Tangent or slopes of polar curves :-

If $r = f(\theta)$, " θ " is parameter and its parametric eq.'s are:

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$y' = \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

9. Find an eq. of T.L to the curve $r = 1 + \sin \theta$, $\theta = \frac{\pi}{3}$

Sol:

$$y - y_0 = m(x - x_0)$$

$$r = 1 + \sin \theta \Big|_{\theta=\frac{\pi}{3}} = \frac{2+\sqrt{3}}{2}$$

$$\frac{dr}{d\theta} = \cos \theta \Big|_{\theta=\frac{\pi}{3}} = \frac{1}{2}$$

$$x_0 = r \cos \theta \Big|_{\theta=\frac{\pi}{3}} = \frac{2+\sqrt{3}}{4}$$

$$y_0 = r \sin \theta \Big|_{\theta=\frac{\pi}{3}} = \frac{2\sqrt{3}+3}{4}$$

$$m = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} = -1$$

$$\therefore y - \frac{2\sqrt{3}+3}{4} = (-1) \left(x - \frac{2+\sqrt{3}}{4} \right)$$

10. Identify the curve by finding cartesian coordinates:

- $r = 2 \sin \theta + 2 \cos \theta$

Sol:

Multiply both sides by “r”

$$r^2 = 2r \sin \theta + 2r \cos \theta$$

$$x^2 + y^2 = 2y + 2x$$

$$x^2 - 2x + y^2 - 2y = 0$$

$$(x - 1)^2 - 1 + (y - 1)^2 - 1 = 0$$

$$(x - 1)^2 + (y - 1)^2 = 2$$

This is circle with center (1,1) and $r = \sqrt{2}$

- $r = \csc \theta \cot \theta$

Sol:

$$r = \frac{\cot \theta}{\sin \theta}$$

$$r \sin \theta = \cot \theta$$

$$y = \frac{x}{y}$$

$$y^2 = x \rightarrow (\text{parabola})$$

10.4-Areas and lengths in Polar coordinates

Suppose there are a curve with polar equation:

$r = f(\theta)$, $a \leq \theta \leq b$. then ,

$$A = \int_a^b \frac{1}{2} r^2 d\theta$$

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Examples

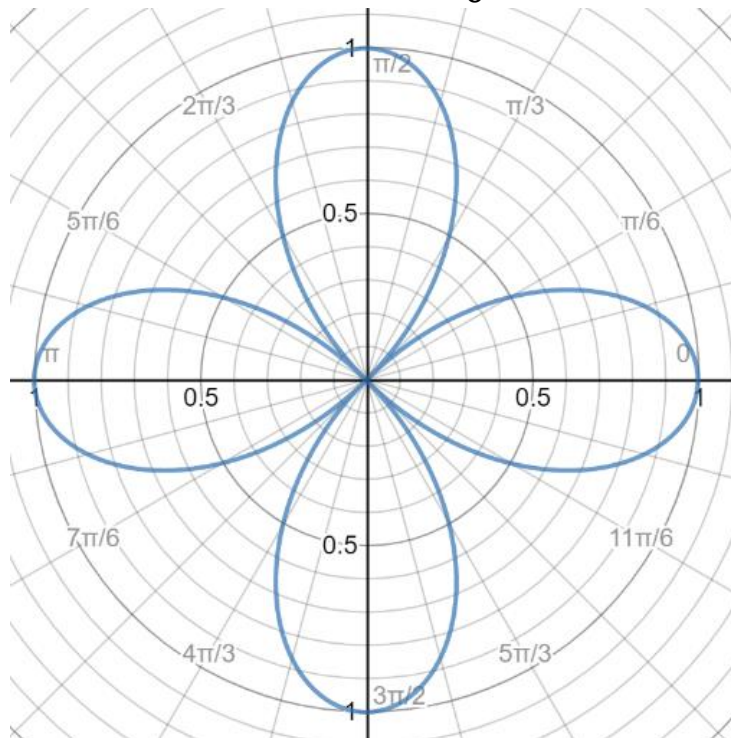
- Find the area enclosed by **one** loop of the 4-leaved rose $r = \cos(2\theta)$.

Sol:

$$r = 0$$

$$\cos(2\theta) = 0 \rightarrow \theta = \pm \frac{\pi}{4}$$

$$\begin{aligned} A &= \int_a^b \frac{1}{2} r^2 d\theta = \frac{1}{2} \cdot 2 \int_0^{\frac{\pi}{4}} (\cos(2\theta))^2 d\theta = \int_0^{\frac{\pi}{4}} \frac{1 + \cos(4\theta)}{2} d\theta \\ &= \frac{1}{2} \left[\theta + \frac{\sin(4\theta)}{4} \right]_0^{\frac{\pi}{4}} = \pi/6 \end{aligned}$$



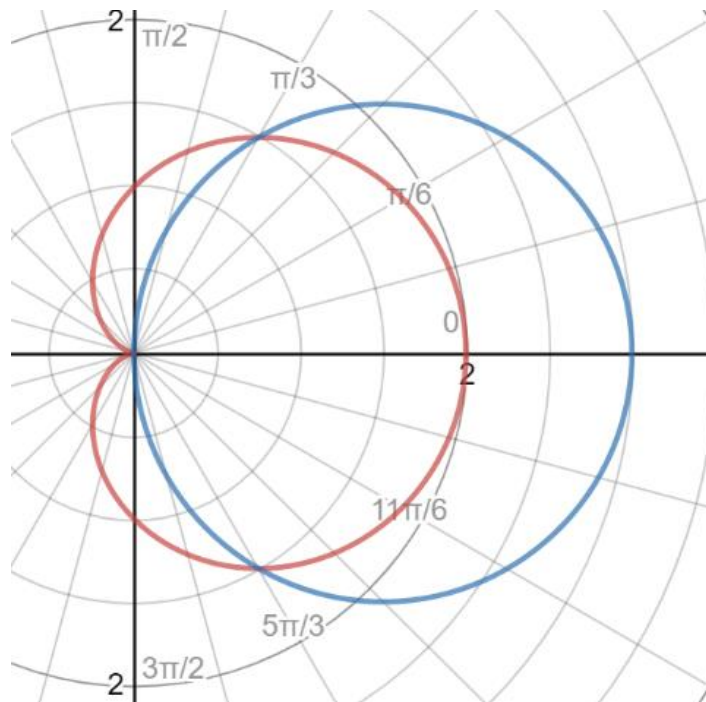
2. Find the area of the region that lies

- A) Inside $r = 3 \cos \theta$ and outside $r = 1 + \cos \theta$.

$$r = r$$

$$3 \cos \theta = 1 + \cos \theta \rightarrow \cos \theta = \frac{1}{2} \rightarrow \theta = \pm \frac{\pi}{3}$$

$$A = \int_a^b \frac{1}{2} r^2 d\theta = \frac{1}{2} \cdot 2 \int_0^{\frac{\pi}{3}} [(3 \cos \theta)^2 - (1 + \cos \theta)^2] d\theta = \pi$$



- B) Inside $r = 1 + \cos \theta$ outside $r = 3 \cos \theta$.

$$\begin{aligned} A &= \int_a^b \frac{1}{2} r^2 d\theta \\ &= \frac{1}{2} \cdot 2 \left[\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} [(1 + \cos \theta)^2 - (3 \cos \theta)^2] d\theta \right. \\ &\quad \left. + \int_{\frac{\pi}{2}}^{\pi} (1 + \cos \theta)^2 d\theta \right] = \frac{\pi}{4} \end{aligned}$$

3. Set up but do not evaluate an integral for the area of the region inside both curves $r = \sin \theta$, $r = \cos \theta$.

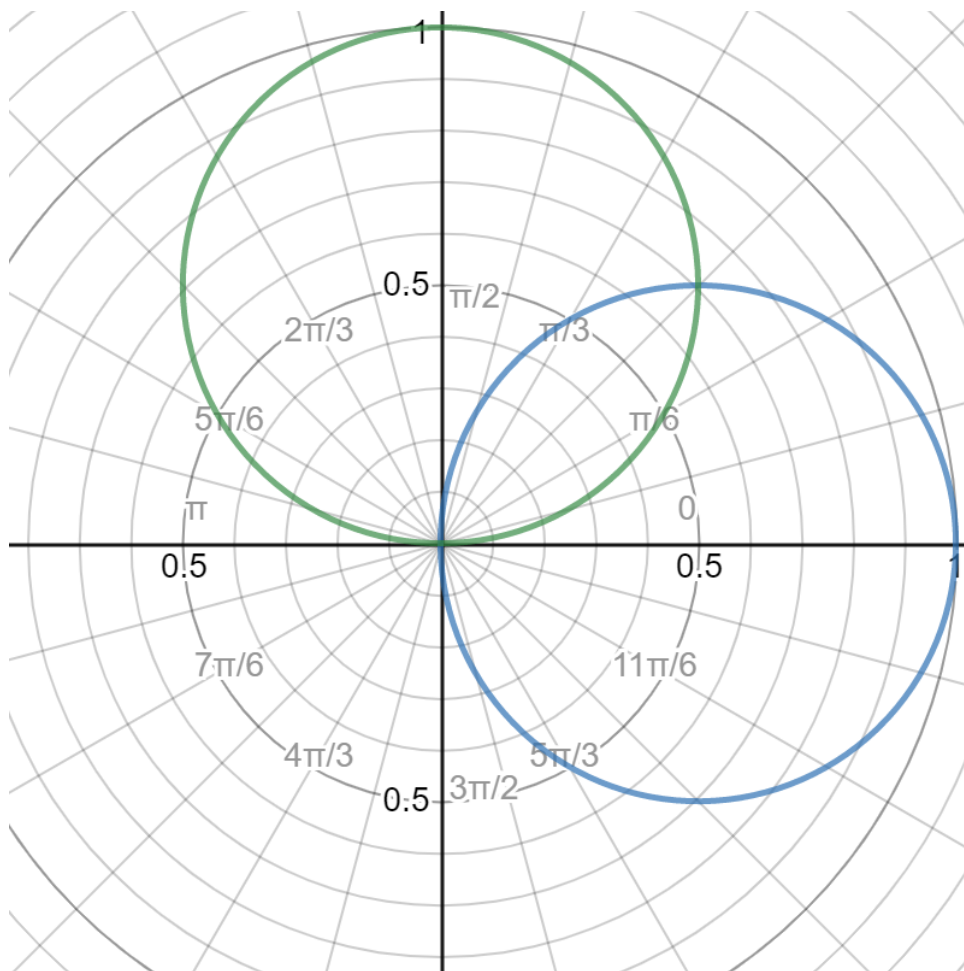
Sol:

$$r = r$$

$$\sin \theta = \cos \theta \rightarrow \theta = \frac{\pi}{4}$$

$$A = \int_a^b \frac{1}{2} r^2 d\theta$$

$$= \frac{1}{2} \cdot 2 \left[\int_0^{\frac{\pi}{4}} (\sin \theta)^2 d\theta = (or) \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cos \theta)^2 d\theta \right] = \frac{\pi}{8} - \frac{1}{4}$$



4. $r = \sin(2\theta)$, $r = \sin \theta$, inside both curves.

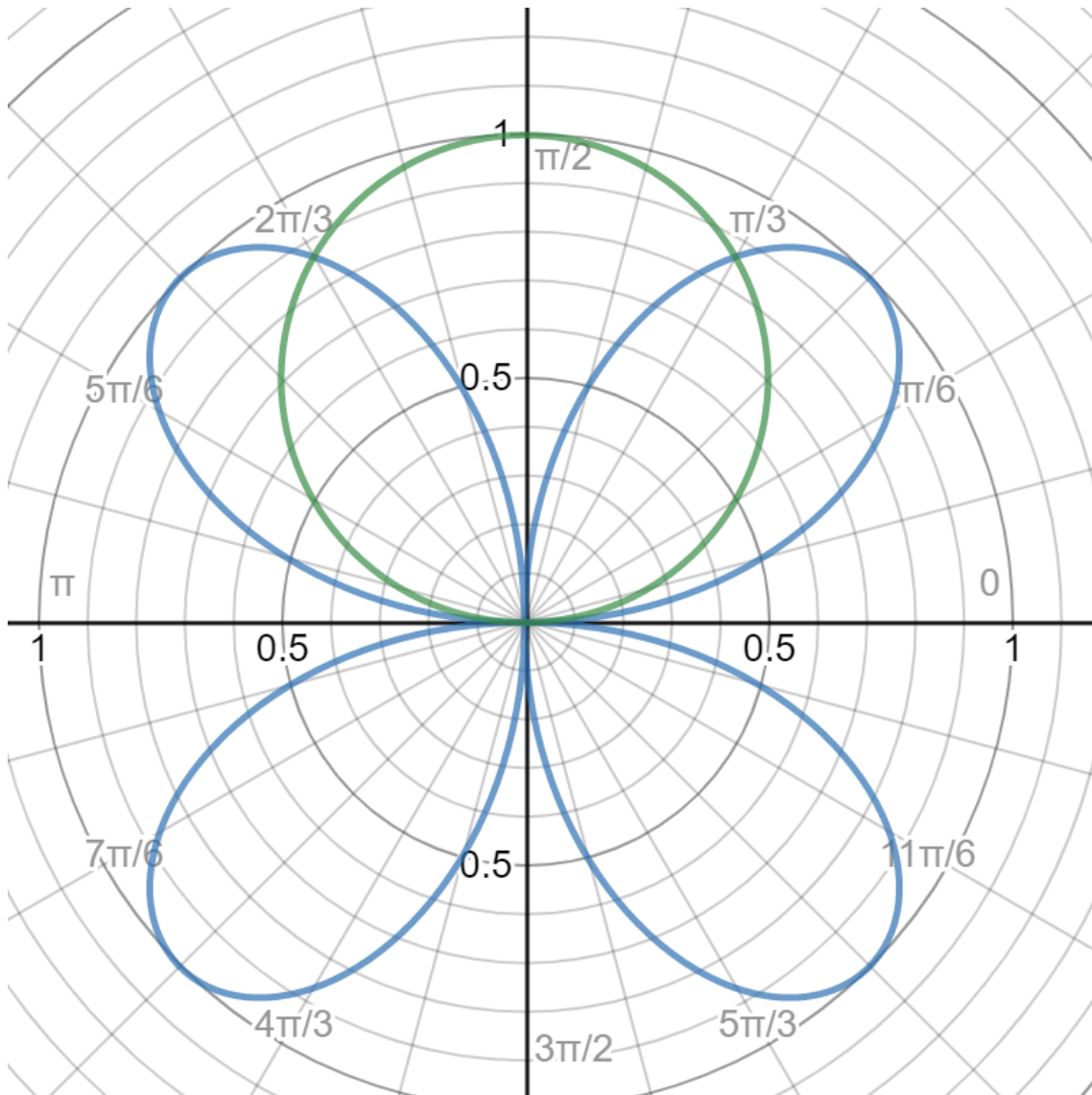
$$r = r$$

$$\sin(2\theta) = \sin \theta \rightarrow 2 \sin \theta \cos \theta = \sin \theta$$

$$\sin \theta (2 \cos \theta - 1) = 0 \rightarrow \theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}$$

$$A = \int_a^b \frac{1}{2} r^2 d\theta$$

$$= \frac{1}{2} \cdot 2 \left[\int_0^{\frac{\pi}{3}} (\sin \theta)^2 d\theta + \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} (\sin(2\theta))^2 d\theta \right]$$



5. Find the length of the polar curve

$$r = \sin \theta + \cos \theta \quad , \quad 0 \leq \theta \leq 2\pi$$

sol:

$$\begin{aligned} L &= \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \\ &= \int_0^{2\pi} \sqrt{(\sin \theta + \cos \theta)^2 + (\cos \theta - \sin \theta)^2} d\theta \\ &= \int_0^{2\pi} \sqrt{\sin^2 \theta + \cos^2 \theta + \sin^2 \theta + \cos^2 \theta} d\theta \\ &= \int_0^{2\pi} \sqrt{2} d\theta \\ &= \sqrt{2} [\theta]_0^{2\pi} = 2\sqrt{2} \pi \end{aligned}$$

6. Find the length of the cardioid $r = 1 + \sin \theta$, $0 \leq \theta \leq 2\pi$

$$\begin{aligned} L &= \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \\ &= \int_0^{2\pi} \sqrt{(1 + \sin \theta)^2 + (\cos \theta)^2} d\theta \\ &= \int_0^{2\pi} \sqrt{1 + 2 \sin \theta + \sin^2 \theta + \cos^2 \theta} d\theta \\ &= \int_0^{2\pi} \sqrt{2 + 2 \sin \theta} d\theta \\ &= \sqrt{2} \int_0^{2\pi} \sqrt{1 + \sin \theta} \frac{\sqrt{1 - \sin \theta}}{\sqrt{1 - \sin \theta}} d\theta \\ &= \sqrt{2} \int_0^{2\pi} \frac{\sqrt{1 - \sin^2 \theta}}{\sqrt{1 - \sin \theta}} d\theta \\ &= \sqrt{2} \int_0^{2\pi} \frac{|\cos \theta|}{\sqrt{1 + \sin \theta}} d\theta \\ &= \sqrt{2} \left[\int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sqrt{1 + \sin \theta}} d\theta + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{-\cos \theta}{\sqrt{1 + \sin \theta}} d\theta + \int_{\frac{3\pi}{2}}^{2\pi} \frac{\cos \theta}{\sqrt{1 + \sin \theta}} d\theta \right] \end{aligned}$$

7. Given $C_1: r = 3(1 + \cos \theta)$, $C_2: r = 4 + \cos \theta$, find area :
- Outside C_1 and inside C_2 .

Sol:

$$A = \frac{1}{2} \cdot 2 \int_{\frac{\pi}{3}}^{\pi} [(4 + \cos \theta)^2 - (3 + 3 \cos \theta)^2] d\theta$$

- Inside C_1 and outside C_2 .

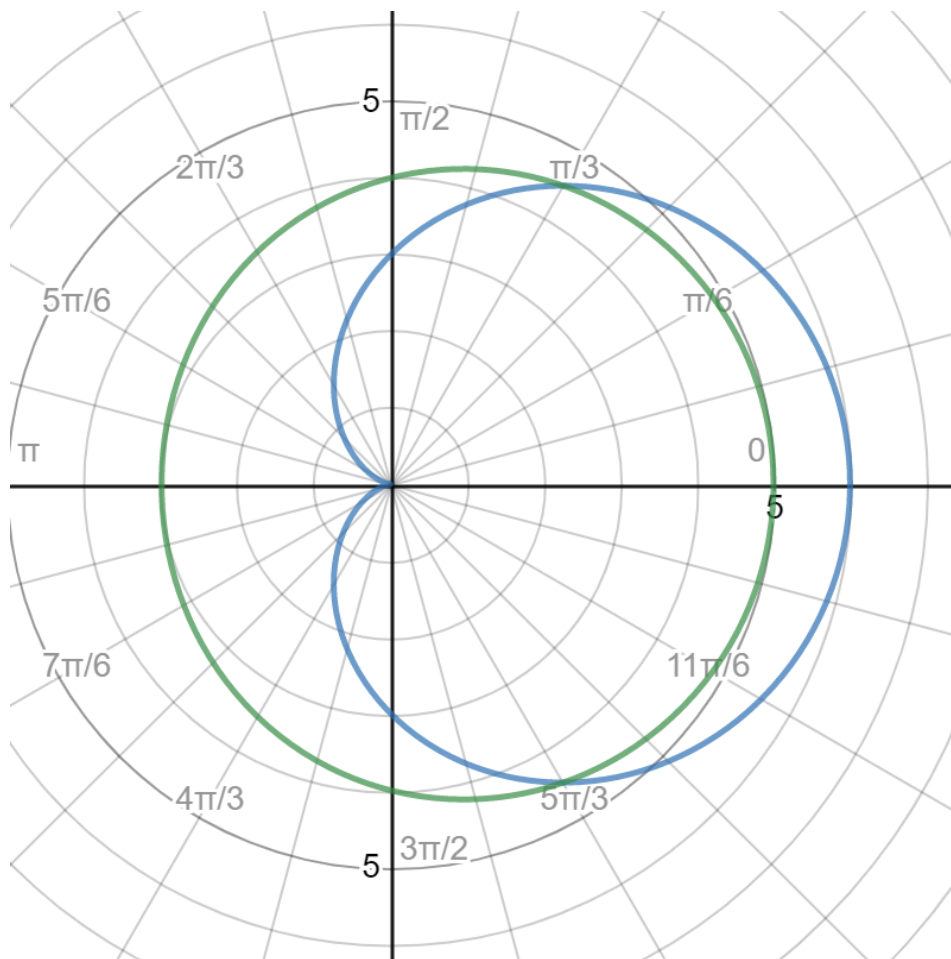
Sol:

$$A = \frac{1}{2} \cdot 2 \int_0^{\frac{\pi}{3}} [(3 + 3 \cos \theta)^2 - (4 + \cos \theta)^2] d\theta$$

- Inside both of them .

- Sol:

$$A = \frac{1}{2} \cdot 2 \left[\int_0^{\frac{\pi}{3}} (4 + \cos \theta)^2 d\theta + \int_{\frac{\pi}{3}}^{\pi} (3 + 3 \cos \theta)^2 d\theta \right]$$



Exercises-10.3 + 10.4

1. Sketch the graph of $r = \sqrt{4 + 5 \cos^2 \theta}$, $0 \leq \theta \leq 2\pi$ by labeling at least 4 points on it and find the area of the region bounded by the graph ?
2. Sketch the graph of $r = \frac{2\theta}{\pi} + 1$, $0 \leq \theta \leq 2\pi$ by labeling at least 3 points on it and find the area of the region bounded by the graph in the second quadrant ?
3. Given $r = e^{3\theta}$
 - Sketch the graph of the polar equation ?
 - Find the slope of the T.L to the above graph at the point that corresponding $\theta = \frac{\pi}{2}$
4. Find the length of the curve $r = 4 \cos^2 \left(\frac{\theta}{2} \right) \rightarrow (ans = 16)$
5. Find the area of the region inside both $r = \sin \theta$
 $r = \sin(2\theta) \rightarrow (ans = \frac{\pi}{4} - \frac{3\pi}{16})$
6. Find the area enclosed by one loop of the curve $r = \sin(3\theta)$
 $\rightarrow (ans = \frac{\pi}{12})$
7. Let $r = 2 + 2 \cos \theta$
 - Find the slope of the T.L at $\theta = \frac{\pi}{4}$
 - Find the eq. of the T.L at $\theta = \frac{\pi}{4}$ (in rectangular coordinates)
8. Find the length of the cardioid $r = 1 + \sin \theta \rightarrow (ans = 8)$